

FM-GPT Tutorial

Data Generation

We will first generate a synthetic dataset using the included functions. For this example, we will consider a set of 10 continuous outcomes which can be represented by 3 factors. We will generate $n_1 = 100$ samples in the eQTL reference dataset and $n_2 = 1000$ samples in the GWAS dataset. We will simulate a singular region of 10 genes, each gene containing 10 cis-SNPs. The LD matrix for the SNPs will be $\rho_{\text{within}} = 0.66$ and $\rho_{\text{between}} = 0.33$. We will assume approximately 1 gene is causal of the 10 in this region, with a gene expression heritability of 10%. We will also specify the phenotype heritability of approximately 5%.

```
set.seed(1423)
dat <- gen_twas_sets(n_qtl = 100, n_gwas = 5000, n_genes = 10, snps_per_gene = 10, block_sizes = rep(10, 10), rho_between = 0.33, rho_within = 0.66, response_types = c(10, 0, 0), k = 3, maf = 0.24, choice = "genes", sig_rho = 0.66, pi0 = 1, pi1 = 0.1, pi2 = 1, expression_heritability = 0.1, sigma2 = 1, binflate = 0.05)
```

The `dat` object contains the following elements:

```
names(dat)
```

```
## [1] "G_qtl"      "G_gwas"      "E_qtl"      "SNP_coef"   "Yl"         "Yo"
## [7] "Lambda"    "Sigma"      "group"      "A"          "beta"       "r"
## [13] "responses" "U"          "Ucov"      "LD"
```

The most important elements are: `G_qtl` and `G_gwas`, which contain the genotype data for the eQTL and GWAS studies, `E_qtl`, which contains the gene expression data for the eQTL study, `Yl` and `Yo` which contain the latent factors and observed outcome data, `Lambda`, which contains the loading matrix, `beta` which contains the gene causality pattern for the latent factors, and finally `responses`, which categorizes the response type of the outcome data (e.g. continuous, binary, count).

Now let's view the gene causality pattern.

```
dat$beta
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.0000000 0.0000000 0.0000000
## [2,] 0.0000000 0.0000000 0.0000000
## [3,] 0.1715735 0.3307978 0.1299771
## [4,] 0.0000000 0.0000000 0.0000000
## [5,] 0.0000000 0.0000000 0.0000000
## [6,] 0.0000000 0.0000000 0.0000000
## [7,] 0.0000000 0.0000000 0.0000000
## [8,] 0.0000000 0.0000000 0.0000000
## [9,] 0.0000000 0.0000000 0.0000000
## [10,] 0.0000000 0.0000000 0.0000000
```

Computation

Now we shall estimate the predicted expression from the synthetic data. We will use a spike-and-slab regression implementation.

```
est_coef_mat <- matrix(nrow = nrow(dat$SNP_coef), ncol = ncol(dat$SNP_coef))

for(j in 1:ncol(dat$SNP_coef)) {
  tmp_out <- lm.spike(scale(dat$E_qtl[,j], scale = FALSE) ~ scale(dat$G_qtl, scale =
FALSE)+0, niter = 1000)
  est_coef_mat[,j] <- apply(tmp_out$beta[751:1000,], 2, median)
}
```

For example, for the first gene, here is the weight vector for its 10 SNPs:

```
est_coef_mat[1:10,1]
```

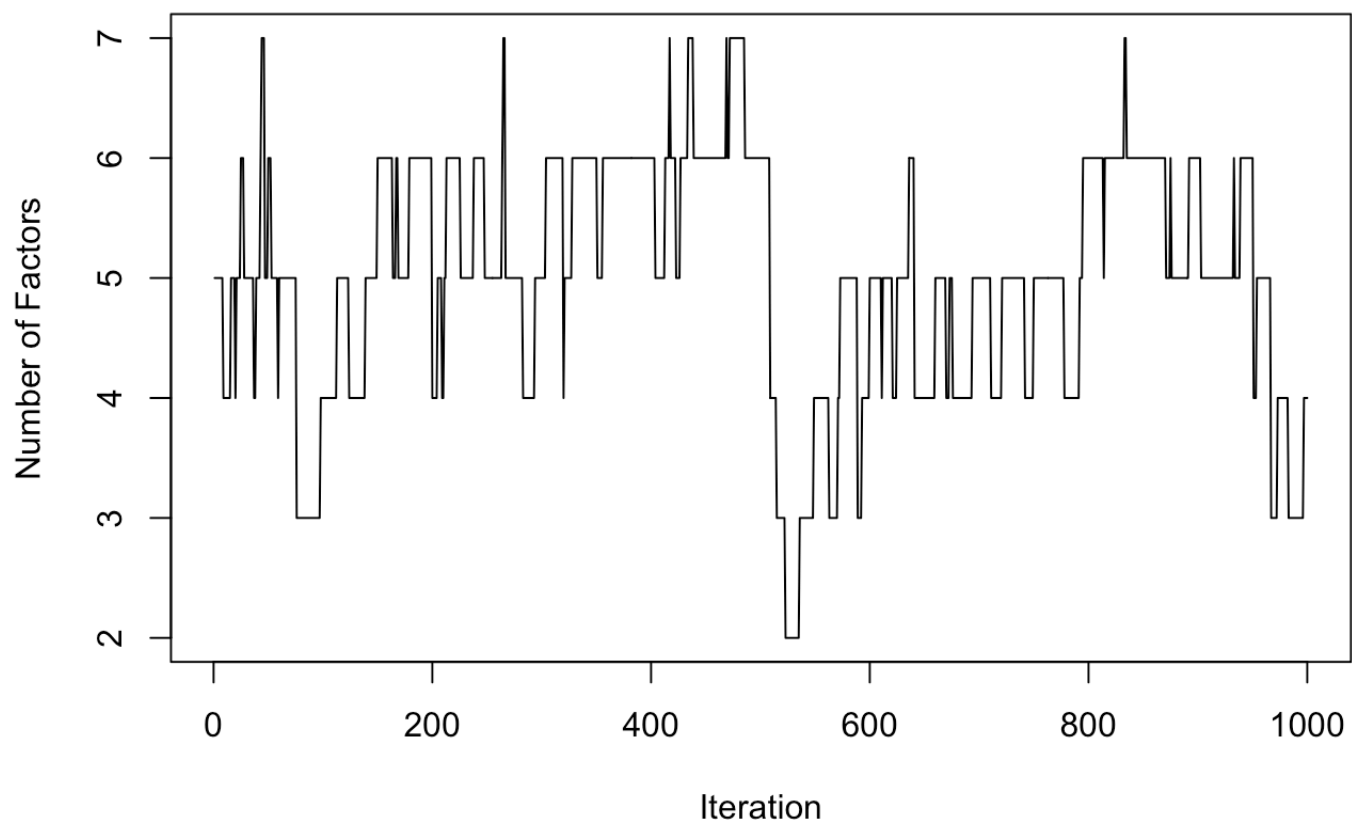
```
## [1] 0.000000 0.000000 0.000000 0.000000 -1.719266 0.000000 0.000000
## [8] 0.000000 0.000000 -1.101098
```

Now, we will generate the predicted gene expressions and fit the model.

```
E_gwas <- dat$G_gwas %*% est_coef_mat
fit <- FMGPT(X = scale(E_gwas, scale = FALSE), Y = scale(dat$Yo, scale = FALSE), A =
0*dat$A, group = seq(1, 10), response_types = dat$responses, epsilon = 0.4, B = 3000,
burnin = 2000, thin = 1, k_fix = NULL, scale.eta = FALSE)
```

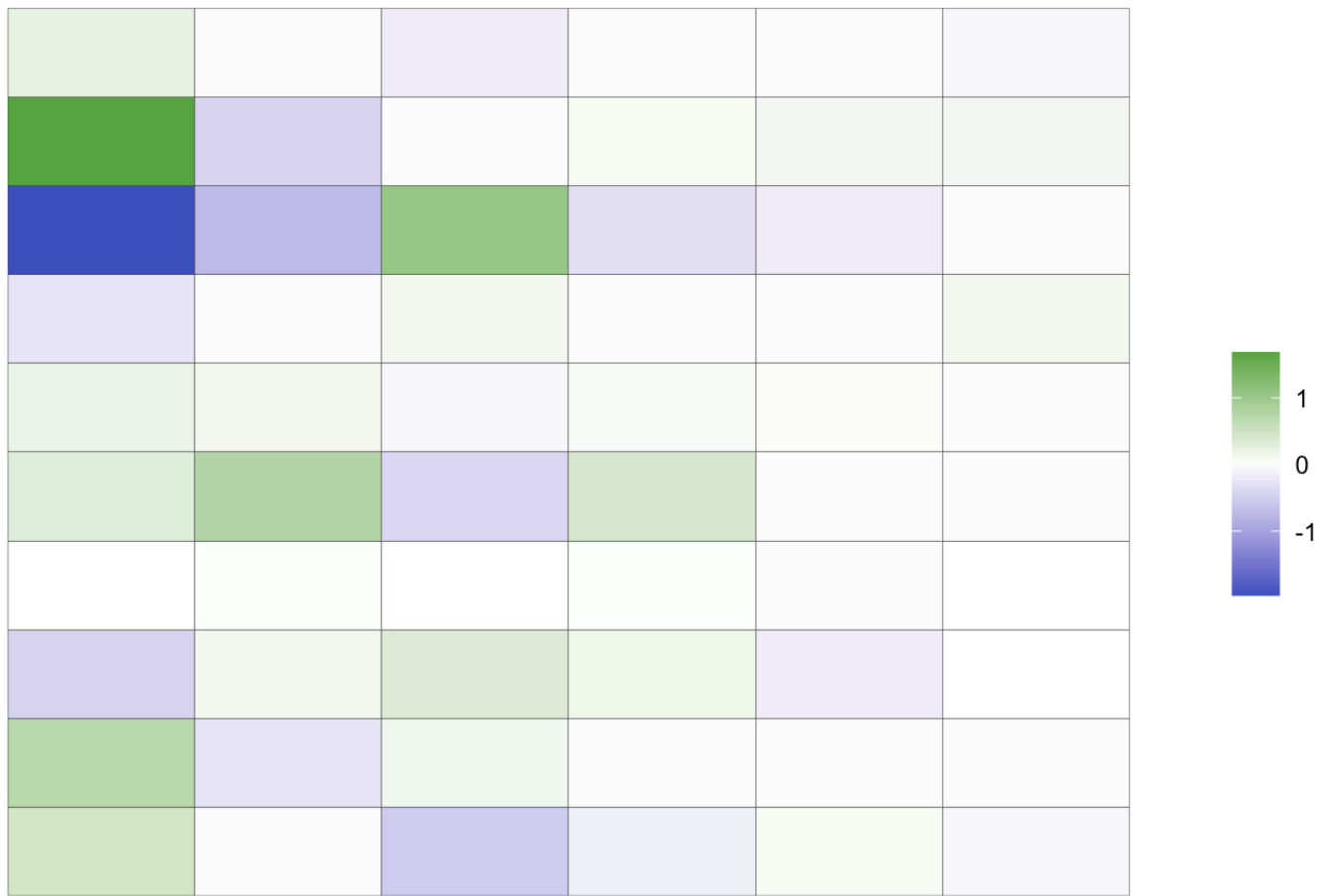
We can first assess the total number of factors the model has suggested exist in the data:

```
ts.plot(fit$kstar, xlab = "Iteration", ylab = "Number of Factors")
```



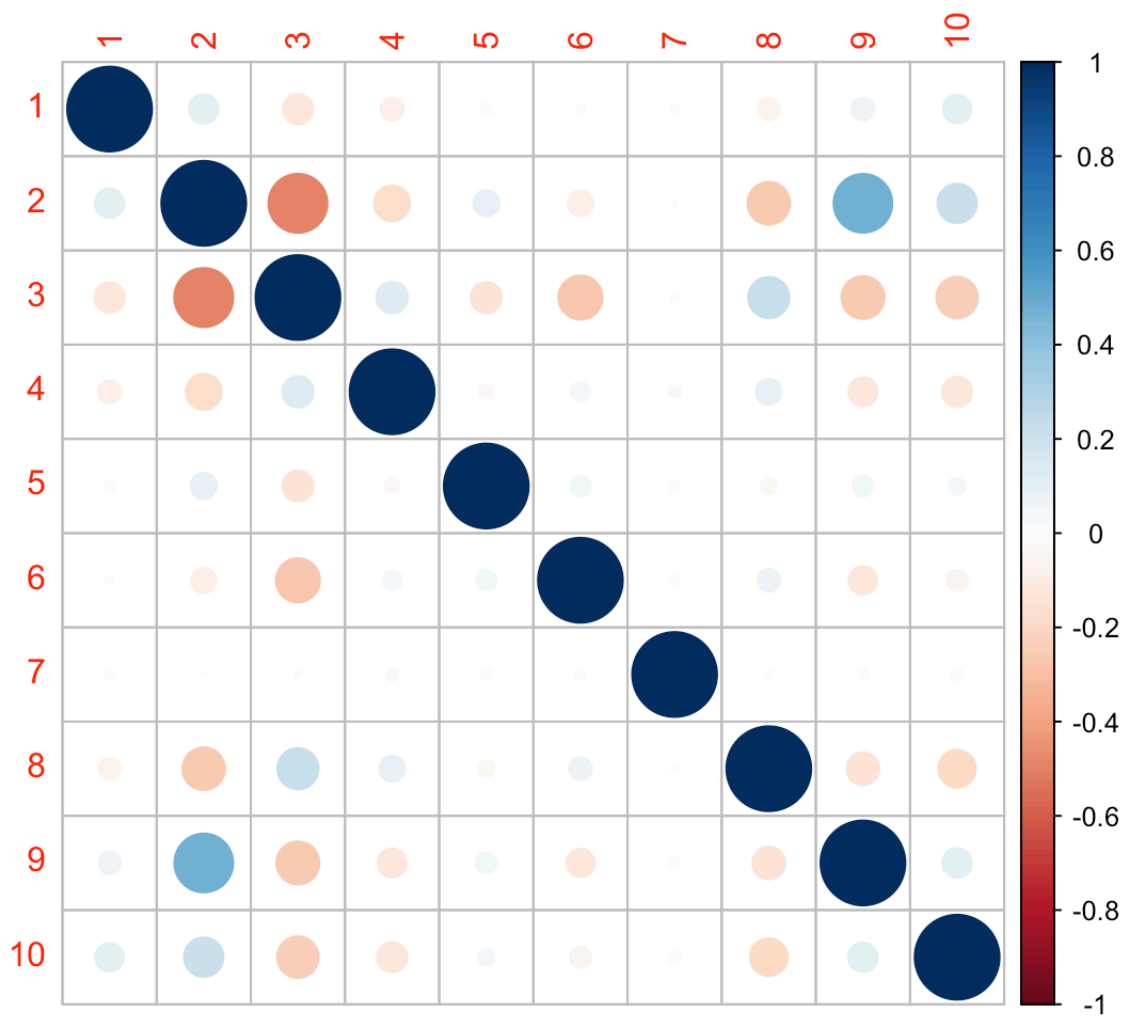
Additionally, we can view the loading matrix pattern. Based on the trace plot of the number of factors, we will view the loadings for the first six factors:

```
infinitefactor::plotmat(t(apply(fit$lambda, c(2, 3), mean, na.rm = TRUE))[,1:6]))
```



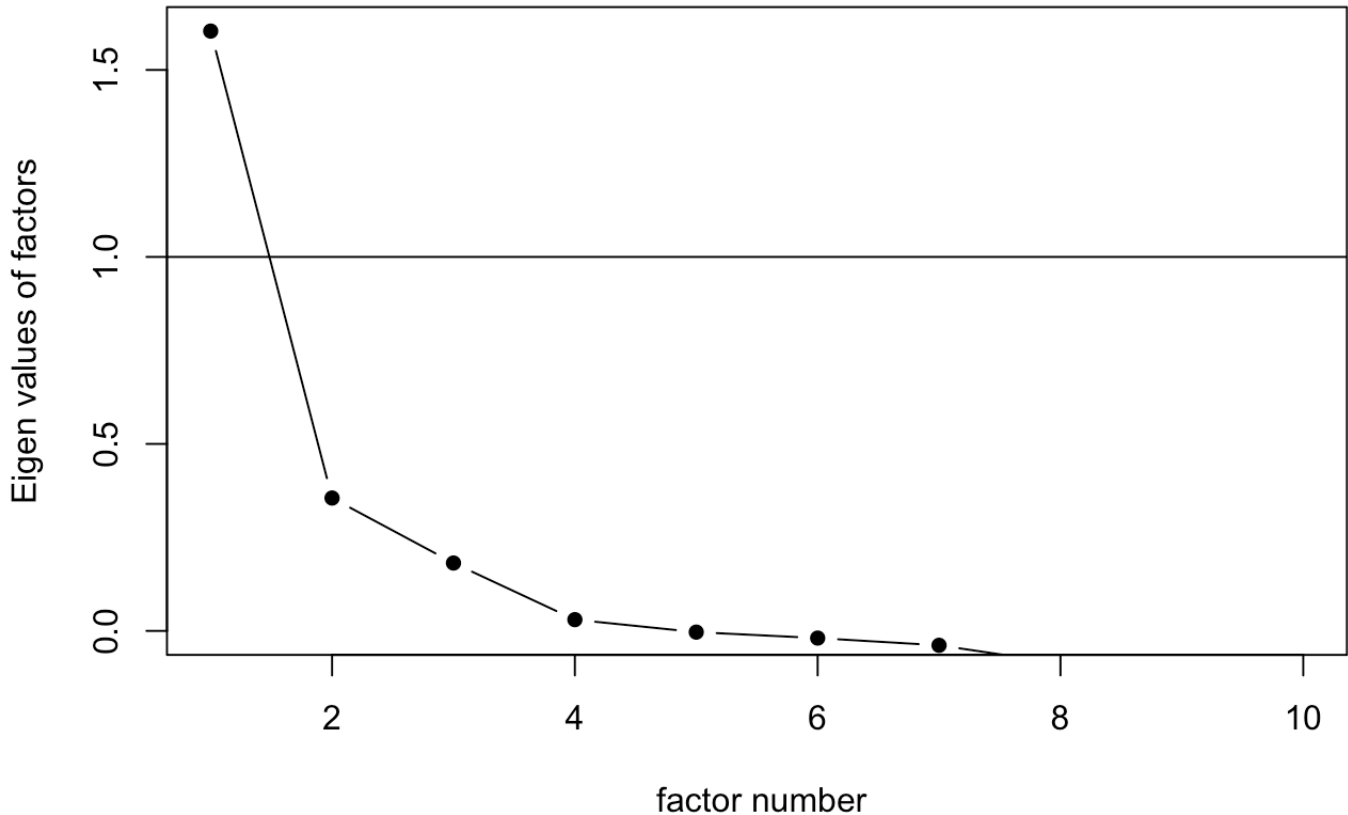
We can also do more classic methods of factor determination, such as the correlation plot and the scree plot:

```
corrplot::corrplot(cor(dat$Yo))
```



```
psych::scree(cor(dat$Yo), pc = FALSE)
```

Scree plot



Furthermore, we can try to suppress certain loadings to zero. Looking at the posterior mean of the loading matrix for the first three factors, we have:

```
t(apply(fit$lambda, c(2, 3), mean, na.rm = TRUE))[,1:3]
```

```
##           [,1]      [,2]      [,3]
## [1,]  0.228136006 -0.021832330 -0.188681266
## [2,]  1.678967752 -0.421212221 -0.026450774
## [3,] -1.975485138 -0.720422383  1.050560210
## [4,] -0.275308863  0.027476766  0.112662100
## [5,]  0.208116483  0.127676866 -0.068475715
## [6,]  0.312933988  0.765620239 -0.387355095
## [7,] -0.003455489  0.009329259  0.002512951
## [8,] -0.447556340  0.106884653  0.345457250
## [9,]  0.719770599 -0.270749420  0.131224191
## [10,] 0.471726854 -0.027442417 -0.529810597
```

Now, we'll apply the post-hoc unit scaling adjustment to determine which loadings to keep within each factor.

```
dim(fit$lambda)
```

```
## [1] 1000 10 10
```

```
lambda_draws <- fit$lambda[,1:3,]  
for(b in 1:750) {  
  for(k in 1:3) {  
    lambda_draws[b,k,] <- lambda_draws[b,k,] / sqrt(sum(lambda_draws[b,k,]^2))  
  }  
}  
t(apply(lambda_draws, c(2, 3), mean, na.rm = TRUE))[,1:3]
```

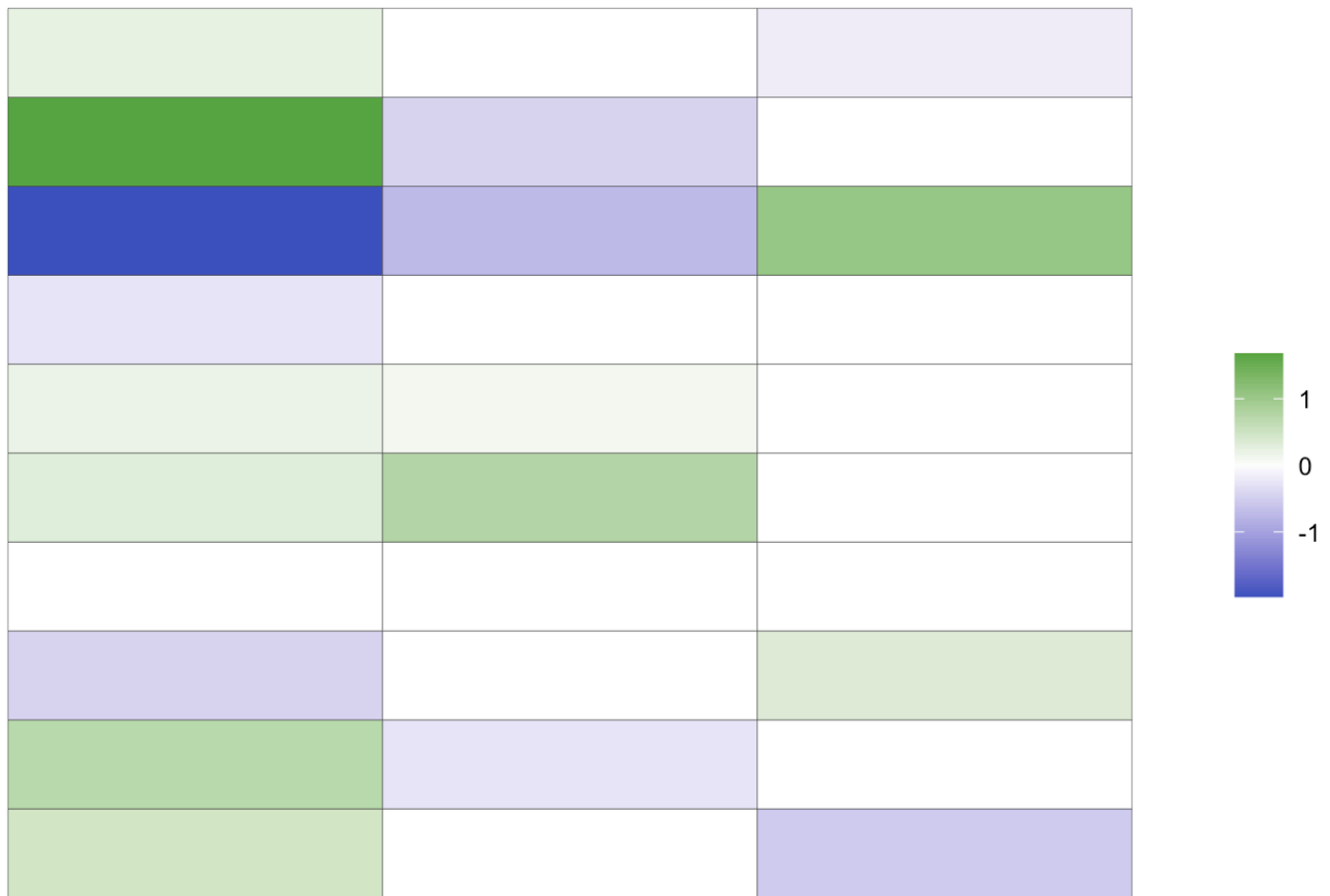
```
##           [,1]           [,2]           [,3]  
## [1,]  0.125128803 -0.0237267830 -0.108230296  
## [2,]  0.871217714 -0.3160903453 -0.037762536  
## [3,] -1.103193680 -0.5374466405  0.314718766  
## [4,] -0.142874434  0.0007746511  0.073747340  
## [5,]  0.106915864  0.1040648931  0.005468073  
## [6,]  0.193282812  0.5339733095  0.071825144  
## [7,] -0.004489443 -0.0034926657  0.018296691  
## [8,] -0.256985029  0.0552749793  0.273067655  
## [9,]  0.365444884 -0.2025981749  0.021096815  
## [10,] 0.267596961 -0.0043584861 -0.364186470
```

This matrix gives the relative magnitude of each loading in each factor. We'll keep only those loadings with an absolute magnitude > 0.10.

```
lambda_keep <- t(apply(lambda_draws, c(2, 3), mean, na.rm = TRUE))[,1:3]  
lambda_keep[abs(lambda_keep) < 0.1] <- 0  
t(apply(fit$lambda, c(2, 3), mean, na.rm = TRUE))[,1:3] * (abs(lambda_keep) > 0.10)
```

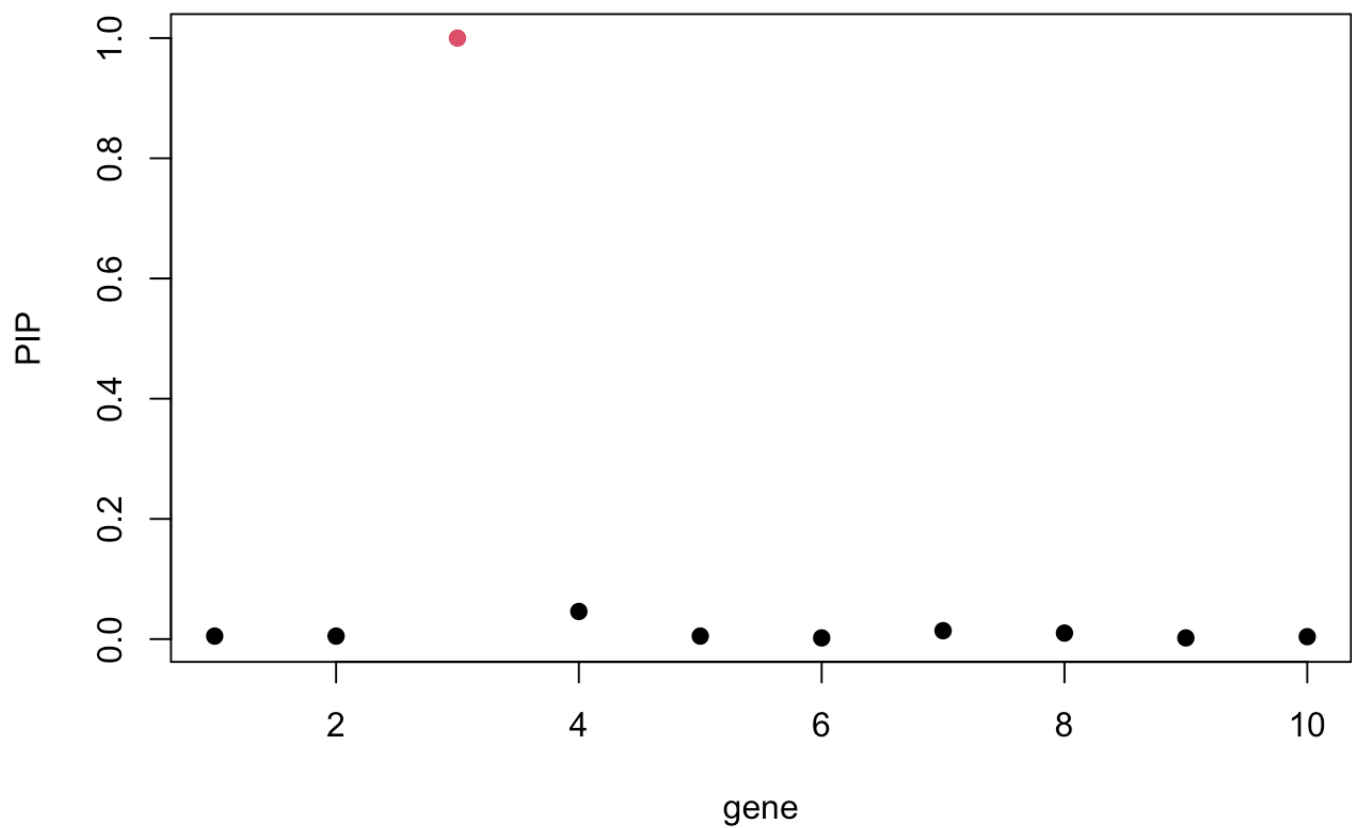
```
##           [,1]           [,2]           [,3]  
## [1,]  0.2281360  0.00000000 -0.1886813  
## [2,]  1.6789678 -0.4212122  0.0000000  
## [3,] -1.9754851 -0.7204224  1.0505602  
## [4,] -0.2753089  0.00000000  0.0000000  
## [5,]  0.2081165  0.1276769  0.0000000  
## [6,]  0.3129340  0.7656202  0.0000000  
## [7,]  0.0000000  0.0000000  0.0000000  
## [8,] -0.4475563  0.0000000  0.3454572  
## [9,]  0.7197706 -0.2707494  0.0000000  
## [10,] 0.4717269  0.0000000 -0.5298106
```

```
infinitefactor::plotmat(t(apply(fit$lambda, c(2, 3), mean, na.rm = TRUE))[,1:3] * (abs(lambda_keep) > 0.10))
```



Finally, now that we've determined to keep 3 factors, we can look at the omnibus PIP of the genes for these three factors:

```
plot(twas_pip(fit, 3), xlab = "gene", ylab= "PIP", col = dat$A+1, pch = 19)
```



Clearly the gene with the highest PIP is the 3rd gene, which matches the gene causality pattern shown above.